

✔ JavaScript Array – ALL METHODS (Complete JSON)

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{
  "arrayExample": "let arr = [1, 2, 3, 4, 5];",

  "methods": [
    {
      "name": "push",
      "definition": "Array ke end me element add karta hai",
      "code": "let arr=[1,2]; arr.push(3); console.log(arr); // [1,2,3]"
    },
    {
      "name": "pop",
      "definition": "Array ke last element ko remove karta hai",
      "code": "let arr=[1,2,3]; arr.pop(); console.log(arr); // [1,2]"
    },
    {
      "name": "shift",
      "definition": "Array ka first element remove karta hai",
      "code": "let arr=[1,2,3]; arr.shift(); console.log(arr); // [2,3]"
    },
    {
      "name": "unshift",
      "definition": "Array ke start me element add karta hai",
      "code": "let arr=[2,3]; arr.unshift(1); console.log(arr); // [1,2,3]"
    },
    {
      "name": "length",
      "definition": "Array ki length batata hai",
      "code": "let arr=[1,2,3]; console.log(arr.length); // 3"
    },
    {
      "name": "concat",
      "definition": "Multiple arrays ko jodta hai",
      "code": "let a=[1]; let b=[2]; console.log(a.concat(b)); // [1,2]"
    },
    {
      "name": "join",
      "definition": "Array ko string me convert karta hai",
      "code": "let arr=[1,2,3]; console.log(arr.join('-')); // 1-2-3"
    },
    {
      "name": "toString",
      "definition": "Array ko string banata hai",
      "code": "let arr=[1,2,3]; console.log(arr.toString()); // 1,2,3"
    },
    {
      "name": "slice",
      "definition": "Array ka part copy karta hai",
      "code": "let arr=[1,2,3,4]; console.log(arr.slice(1,3)); // [2,3]"
    },
    {
      "name": "splice",
      "definition": "Array me add/remove/replace karta hai",

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        "code": "let arr=[1,2,3]; arr.splice(1,1,9); console.log(arr); //
[1,9,3]"
    },
    {
        "name": "indexOf",
        "definition": "Element ka index deta hai",
        "code": "let arr=[10,20,30]; console.log(arr.indexOf(20)); // 1"
    },
    {
        "name": "lastIndexOf",
        "definition": "Last matching element ka index deta hai",
        "code": "let arr=[1,2,1]; console.log(arr.lastIndexOf(1)); // 2"
    },
    {
        "name": "includes",
        "definition": "Check karta hai value exist karti hai ya nahi",
        "code": "let arr=[1,2,3]; console.log(arr.includes(2)); // true"
    },
    {
        "name": "map",
        "definition": "Har element par operation karke new array deta hai",
        "code": "let arr=[1,2]; console.log(arr.map(x=>x*2)); // [2,4]"
    },
    {
        "name": "filter",
        "definition": "Condition match hone wale elements return karta hai",
        "code": "let arr=[1,2,3,4]; console.log(arr.filter(x=>x>2)); // [3,4]"
    },
    {
        "name": "reduce",
        "definition": "Array ko single value me convert karta hai",
        "code": "let arr=[1,2,3]; console.log(arr.reduce((a,b)=>a+b,0)); // 6"
    },
    {
        "name": "forEach",
        "definition": "Har element par loop chalata hai",
        "code": "let arr=[1,2]; arr.forEach(x=>console.log(x));"
    },
    {
        "name": "find",
        "definition": "First matching element return karta hai",
        "code": "let arr=[1,2,3]; console.log(arr.find(x=>x>1)); // 2"
    },
    {
        "name": "findIndex",
        "definition": "First matching element ka index deta hai",
        "code": "let arr=[1,2,3]; console.log(arr.findIndex(x=>x>1)); // 1"
    },
    {
        "name": "some",
        "definition": "Koi ek element condition match kare to true",
        "code": "let arr=[1,2,3]; console.log(arr.some(x=>x>2)); // true"
    },
    {
        "name": "every",
        "definition": "Sab element condition match kare to true",
        "code": "let arr=[1,2,3]; console.log(arr.every(x=>x>0)); // true"
    }

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    },
    {
      "name": "sort",
      "definition": "Array ko sort karta hai",
      "code": "let arr=[3,1,2]; arr.sort((a,b)=>a-b); console.log(arr); //
[1,2,3]"
    },
    {
      "name": "reverse",
      "definition": "Array ka order reverse karta hai",
      "code": "let arr=[1,2,3]; arr.reverse(); console.log(arr); // [3,2,1]"
    },
    {
      "name": "fill",
      "definition": "Array ko same value se fill karta hai",
      "code": "let arr=[1,2,3]; arr.fill(0); console.log(arr); // [0,0,0]"
    },
    {
      "name": "copyWithin",
      "definition": "Array ke andar hi copy karta hai",
      "code": "let arr=[1,2,3,4]; arr.copyWithin(1,0); console.log(arr);"
    },
    {
      "name": "flat",
      "definition": "Nested array ko single array banata hai",
      "code": "let arr=[1,[2,[3]]]; console.log(arr.flat(2)); // [1,2,3]"
    },
    {
      "name": "flatMap",
      "definition": "map + flat ek sath karta hai",
      "code": "let arr=[1,2]; console.log(arr.flatMap(x=>[x,x*2]));"
    },
    {
      "name": "entries",
      "definition": "Index + value ka iterator deta hai",
      "code": "let arr=[1,2]; for(let [i,v] of
arr.entries()){console.log(i,v);}"
    },
    {
      "name": "keys",
      "definition": "Array ke index return karta hai",
      "code": "let arr=[1,2]; for(let k of arr.keys()){console.log(k);}"
    },
    {
      "name": "values",
      "definition": "Array ke values return karta hai",
      "code": "let arr=[1,2]; for(let v of arr.values()){console.log(v);}"
    },
    {
      "name": "Array.isArray",
      "definition": "Check karta hai value array hai ya nahi",
      "code": "console.log(Array.isArray([1,2])); // true"
    },
    {
      "name": "at",
      "definition": "Index (negative bhi) se element return karta hai",
      "code": "let arr=[10,20,30]; console.log(arr.at(-1)); // 30"
    }
  ]

```

✓ express-validator – ALL VALIDATION FUNCTIONS

Mostly use hone wale **check** / **body** / **param** / **query chain methods**

(npm install express-validator)

◆ 1.Field Selectors (kahan se data lena)

Function	Definition	Example
<code>body()</code>	req.body se field validate karta hai	<code>body("email").isEmail()</code>
<code>param()</code>	URL params validate karta hai	<code>param("id").isMongoId()</code>
<code>query()</code>	Query string validate karta hai	<code>query("page").isInt()</code>
<code>header()</code>	Headers validate karta hai	<code>header("token").notEmpty()</code>
<code>cookie()</code>	Cookies validate karta hai	<code>cookie("auth").exists()</code>
<code>check()</code>	Body / Param / Query sab pe kaam karta hai	<code>check("email").isEmail()</code>

◆ 2.Required / Empty Checks

Function	Definition	Example
<code>notEmpty()</code>	Field empty nahi honi chahiye	<code>body("name").notEmpty()</code>
<code>exists()</code>	Field exist honi chahiye	<code>body("email").exists()</code>
<code>optional()</code>	Field optional banata hai	<code>body("gst").optional()</code>
<code>body("name").notEmpty().withMessage("Name is required");</code>		

◆ 3.String Validations

Function	Definition	Example
<code>isString()</code>	String check	<code>body("name").isString()</code>

Function	Definition	Example
<code>isLength()</code>	Length check	<code>body("password").isLength({min:6})</code>
<code>isUppercase()</code>	Uppercase string	<code>body("code").isUppercase()</code>
<code>isLowercase()</code>	Lowercase string	<code>body("email").isLowercase()</code>
<code>matches()</code>	Regex validation	<code>body("pan").matches(/^[A-Z]{5}/)</code>
<code>contains()</code>	Word contain karta hai	<code>body("msg").contains("hello")</code>

◆ 4.Number Validations

Function	Definition	Example
<code>isNumeric()</code>	Sirf number	<code>body("age").isNumeric()</code>
<code>isInt()</code>	Integer check	<code>body("age").isInt()</code>
<code>isFloat()</code>	Decimal number	<code>body("price").isFloat()</code>
<code>isDecimal()</code>	Decimal format	<code>body("amount").isDecimal()</code>
<code>isIn()</code>	Allowed values check	<code>body("role").isIn(["admin", "user"])</code>

◆ 5.Email / Phone / URL

Function	Definition	Example
<code>isEmail()</code>	Email validation	<code>body("email").isEmail()</code>
<code>isMobilePhone()</code>	Phone validation	<code>body("phone").isMobilePhone("en-IN")</code>
<code>isURL()</code>	URL validation	<code>body("website").isURL()</code>

◆ 6.Boolean / Date / JSON

Function	Definition	Example
<code>isBoolean()</code>	Boolean check	<code>body("active").isBoolean()</code>
<code>isDate()</code>	Date format	<code>body("dob").isDate()</code>
<code>isISO8601()</code>	ISO date	<code>body("date").isISO8601()</code>
<code>isJSON()</code>	JSON string	<code>body("data").isJSON()</code>

◆ 7.Password / Security

Function	Definition	Example
<code>isStrongPassword()</code>	Strong password check	<code>body("password").isStrongPassword()</code>
<code>custom()</code>	Custom validation	<code>body("age").custom()</code>
<pre>body("password").isStrongPassword({ minLength: 8, minNumbers: 1, minSymbols: 1 });</pre>		

◆ 8.MongoDB / UUID

Function	Definition	Example
<code>isMongoId()</code>	Mongo ObjectId	<code>param("id").isMongoId()</code>
<code>isUUID()</code>	UUID check	<code>body("uuid").isUUID()</code>

◆ 9.Sanitizers (Data clean karna)

Function	Definition	Example
<code>trim()</code>	Extra spaces remove	<code>body("name").trim()</code>
<code>escape()</code>	XSS protection	<code>body("comment").escape()</code>
<code>toInt()</code>	String → number	<code>body("age").toInt()</code>

Function	Definition	Example
<code>toFloat()</code>	Convert to float	<code>body("price").toFloat()</code>
<code>toLowerCase()</code>	Lowercase	<code>body("email").toLowerCase()</code>
<code>toUpperCase()</code>	Uppercase	<code>body("pan").toUpperCase()</code>